

Motor Vehicle Theft Analysis

1. Project Overview

Developed a Motor Vehicle Theft Analysis project using Python, SQL, and Power BI to identify theft trends, regional patterns, and high-risk vehicle categories. Performed data cleaning in Python, analyzed data using SQL queries, and created an interactive Power BI dashboard to visualize key insights and support data-driven decision-making.

2. Dataset Summary

- Rows: 4,553
- Columns: 15
- Key Features:
 - Stolen_vehicles (vehicle_id, vehicle_type, make_id, model_year, vehicle_desc, color, date_stolen, location_id)
 - Make_details (make_id, make_name, make_type)
 - Locations (location_id, region, country, population, density)

3. Data Cleaning using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `describe()` for summary statistics.

	vehicle_id	vehicle_type	make_id	model_year	vehicle_desc	color	date_stolen	location_id	region
count	4553.000000	4527	4538.000000	4538.00000	4520	4538	4553	4553.000000	4553
unique	NaN	25	NaN	NaN	1177	14	182	NaN	13
top	NaN	Stationwagon	NaN	NaN	HILUX	Silver	4/4/22	NaN	Auckland
freq	NaN	945	NaN	NaN	137	1272	83	NaN	1638
mean	2277.000000	NaN	584.128471	2005.23182	NaN	NaN	NaN	105.779925	NaN
std	1314.482217	NaN	35.062695	9.11763	NaN	NaN	NaN	4.686655	NaN
min	1.000000	NaN	501.000000	1940.00000	NaN	NaN	NaN	101.000000	NaN
25%	1139.000000	NaN	550.000000	2000.00000	NaN	NaN	NaN	102.000000	NaN
50%	2277.000000	NaN	587.000000	2005.00000	NaN	NaN	NaN	104.000000	NaN
75%	3415.000000	NaN	619.000000	2011.00000	NaN	NaN	NaN	109.000000	NaN
max	4553.000000	NaN	638.000000	2022.00000	NaN	NaN	NaN	116.000000	NaN

country	population	density	make_name	make_type
4553	4553	4553.000000	4538	4538
1	13	NaN	138	2
New Zealand	1,695,200	NaN	Toyota	Standard
4553	1638	NaN	716	4348
NaN	NaN	141.373949	NaN	NaN
NaN	NaN	152.664803	NaN	NaN
NaN	NaN	3.280000	NaN	NaN
NaN	NaN	14.720000	NaN	NaN
NaN	NaN	28.800000	NaN	NaN
NaN	NaN	343.090000	NaN	NaN
NaN	NaN	343.090000	NaN	NaN

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4553 entries, 0 to 4552
Data columns (total 14 columns):
#   Column          Non-Null Count  Dtype
---  -
0   vehicle_id      4553 non-null   int64
1   vehicle_type    4527 non-null   object
2   make_id         4538 non-null   float64
3   model_year      4538 non-null   float64
4   vehicle_desc    4520 non-null   object
5   color           4538 non-null   object
6   date_stolen     4553 non-null   object
7   location_id     4553 non-null   int64
8   region          4553 non-null   object
9   country         4553 non-null   object
10  population       4553 non-null   object
11  density          4553 non-null   float64
12  make_name        4538 non-null   object
13  make_type        4538 non-null   object
dtypes: float64(3), int64(2), object(9)
memory usage: 498.1+ KB
```

- **Missing Data Handling:**
 - Drop the rows where non null values are less than 8.
 - Checked for null values and fill the missing values in the **Vehicle_type** column using the mode of vehicle_type column.
 - Fill the missing values in the **Vehicle_desc** column with the word “**unknown**”.
- **Remove Irrelevant Data:** Drop the **country** column.
- **Convert Data Type:** Converting the data type of **Data_stolen** column to 'datetime'.

- **Feature Engineering:**
 - Created a new column **month** by accessing the month from date_stolen column.
 - Created a new column **year** by accessing the year from date_stolen column.
 - Created a new column **week_day** by accessing the day_name from date_stolen column.
 - Created a new column **vehicle_age**.
- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

1. **What types of vehicles are most often and least often stolen? Does this vary by region?**

Most Stolen Vehicle types

	vehicle_type	count(vehicle_id)
▶	Stationwagon	956
	Saloon	851
	Hatchback	644
	Trailer	582
	Utility	466

Least Stolen Vehicle types

	vehicle_type	count(vehicle_id)
▶	Special Purpose Vehicle	1
	Articulated Truck	1
	Trail Bike	2
	Mobile Machine	2
	Tractor	4

Most Stolen Vehicle types By Region

vehicle_type	region	No_of_Vehide_stolen
Stationwagon	Auckland	310
Utility	Bay of Plenty	100
Saloon	Bay of Plenty	86
Stationwagon	Canterbury	165
Trailer	Canterbury	128
Saloon	Gisborne	41
Stationwagon	Gisborne	39
Trailer	Hawke's Bay	23
Saloon	Hawke's Bay	20
Stationwagon	Manawatū-...	30
Saloon	Manawatū-...	28
Trailer	Nelson	19
Stationwagon	Nelson	17
Stationwagon	Northland	70
Saloon	Northland	37
Stationwagon	Otago	33
Saloon	Otago	29
Stationwagon	Southland	9
Saloon	Southland	7
Stationwagon	Taranaki	23
Trailer	Taranaki	21
Saloon	Taranaki	21
Saloon	Waikato	80
Stationwagon	Waikato	72
Stationwagon	Wellington	92
Saloon	Wellington	69

Least Stolen Vehicle types by Region

	vehicle_type	region	No_of_Vehide_stolen
	Mobile Home - Light	Bay of Plenty	1
	Convertible	Bay of Plenty	1
	Trail Bike	Canterbury	1
	Heavy Van	Canterbury	1
	Flat Deck Truck	Gisborne	1
	Other Truck	Gisborne	1
	Trailer - Heavy	Hawke's Bay	1
	Boat Trailer	Hawke's Bay	1
	Caravan	Manawatū-...	1
	Sports Car	Manawatū-...	1
	Heavy Van	Nelson	1
	Trailer - Heavy	Nelson	2
	Cab and Chassis O...	Northland	1
	Flat Deck Truck	Northland	1
	Trail Bike	Otago	1
	Cab and Chassis O...	Otago	1
	Caravan	Southland	1
	Moped	Southland	1
	Light Van	Taranaki	2
	Flat Deck Truck	Taranaki	2
	Heavy Van	Waikato	1
	Articulated Truck	Waikato	1
	Heavy Van	Wellington	1
	Other Truck	Wellington	1

2. In which regions vehicles are most and least stolen?

	region	population	density	No of Vehide_stolen
►	Auckland	1,695,200	343.09	1630
	Canterbury	655,000	14.72	660
	Bay of Plenty	347,700	28.8	445
	Wellington	543,500	67.52	417
	Waikato	513,800	21.5	369
	Northland	201,500	16.11	234
	Gisborne	52,100	6.21	175
	Otago	246,000	7.89	139
	Manawatū-Whanganui	258,200	11.62	139
	Taranaki	127,300	17.55	112
	Hawke's Bay	182,700	12.92	100
	Nelson	54,500	129.15	92
	Southland	102,400	3.28	26

3. Which make type are most frequently stolen?

	make_type	No of Vehicle_stolen
►	Standard	4348
	Luxury	190

By Region

	make_type	region	No of Vehicle_stolen
	Standard	Canterbury	640
	Standard	Bay of Plenty	431
	Standard	Wellington	405
	Standard	Waikato	364
	Standard	Northland	228
	Standard	Gisborne	171
	Standard	Otago	135
	Standard	Manawatū-Whanganui	135
	Luxury	Auckland	114
	Standard	Taranaki	109
	Standard	Hawke's Bay	98
	Standard	Nelson	90
	Standard	Southland	26
	Luxury	Canterbury	20
	Luxury	Bay of Plenty	14
	Luxury	Wellington	12
	Luxury	Northland	6
	Luxury	Waikato	5
	Luxury	Gisborne	4
	Luxury	Otago	4
	Luxury	Manawatū-Whanganui	4
	Luxury	Taranaki	3
	Luxury	Nelson	2
	Luxury	Hawke's Bay	2

4. What are the top 10 most stolen make name?

	make_name	No of Vehicle_stolen
►	Toyota	716
	Trailer	543
	Nissan	482
	Mazda	433
	Ford	312
	Honda	271
	Mitsubishi	204
	Subaru	202
	Suzuki	164
	Holden	160

5. What colors are most and least stolen?

	color	No of Vehicle_stolen
►	Silver	1272
	White	934
	Black	589
	Blue	512
	Red	390
	Grey	378
	Green	224
	Gold	77
	Brown	49
	Yellow	39
	Orange	35
	Purple	26
	Cream	9
	Pink	4

6. Which year have the most and least stolen Vehicles?

	year	No of Vehicle_stolen
►	2021	1663
	2022	2875

7. In which month vehicles are most and least stolen?

	month	year	No of Vehicle_stolen
►	3	2022	1050
	2	2022	758
	1	2022	740
	12	2021	642
	11	2021	559
	10	2021	462
	4	2022	327

8. In which weekday vehicles are most and least stolen?

	week_day	No of Vehicle_stolen
►	Monday	762
	Tuesday	708
	Friday	654
	Wednesday	625
	Thursday	618
	Sunday	595
	Saturday	576

9. Does Vehicle Age effect the theft?

	vehide_age_group	No of Vehicle_stolen
▶	0-15	1870
	15-30	2431
	30-45	192
	45-60	39
	60-75	4
	75-85	2

New Vehicle are more stolen as compare to older.

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.





